

Lecture6

September 9, 2026

1 Lecture 6

September 4, 2026

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[47]: # Modules are toolboxes. You can share your toolbox with other people
      # math, numpy, matplotlib, scipy (useful for scientific computing)
      # tensorflow, pytorch, keras, pandas, opencv (machine learning)

      # Plotting
      import matplotlib.pyplot as plt

      # Numerical Python
      import numpy as np

      # Math
      import math
```

```
[23]: x = math.cos(math.pi)
      print(x)
```

-1.0

```
[27]: # exponential
      print(math.exp(1))
```

2.718281828459045

```
[29]: # natural logarithm
      print(math.log(1))
```

0.0

```
[37]: # linspace
      # The last value is the number of entries. The last value is included.
      ↪ Generates 6 entries from 0 to 2.
      # Equally spaced, divided by the number of entries minus 1.
      v = np.linspace(0, 2, 6)
      print(v)
```

[0. 0.4 0.8 1.2 1.6 2.]

```
[39]: # arange
# The last element is the increment. It looks like the range function, but it's
      ↪ a vector.
u = np.arange(0, 1, 0.2)
print(u)
```

```
[0.  0.2 0.4 0.6 0.8]
```

```
[132]: # Use matplotlib
y1 = np.exp(x) - 1
y2 = x**2

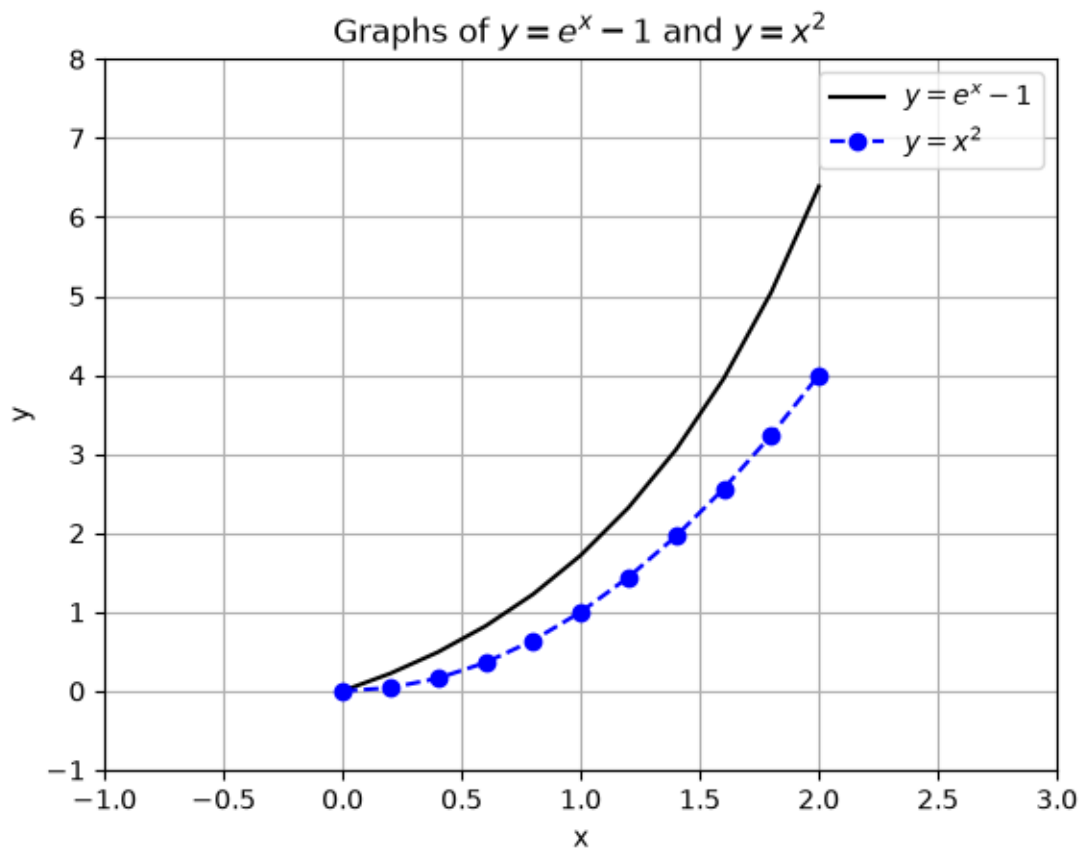
# You put 101 because you have to do (2 - 0) / (101 - 1)
x = np.linspace(0, 2, 11)

# They need to have the same length
# k is black
# o is circle
plt.plot(x, y1, 'k-', label='$y=e^x-1$')
plt.plot(x, y2, 'bo--', label='$y=x^2$')

# Limits
plt.xlim(-1, 3)
plt.ylim(-1, 8)

# Labels
plt.xlabel('x')
plt.ylabel('y')
plt.title('Graphs of $y=e^x-1$ and $y=x^2$')
plt.legend()
plt.grid()

plt.show()
```



```
[ ]: # Practice problem
      # Evaluate P(x)

def polynomial(c, x):
    """
    Input c, a list with the polynomial coefficients,
    and x, the variable to put into the polynomial
    """
    p = 0

    for i in range(0, len(c)):
        p+=c[i]*x**i

    return p

c = [1,2,3]

print(polynomial(c, 100))
```

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[5]: # passing lists into functions
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```
def swap(d, i, j):  
    a = d[i]  
    b = d[j]  
  
    d[j] = a  
    d[i] = b  
  
swapped = [1,2,3,4]  
print(swapped)  
  
swap(swapped, 2, 3)  
  
print(swapped)
```

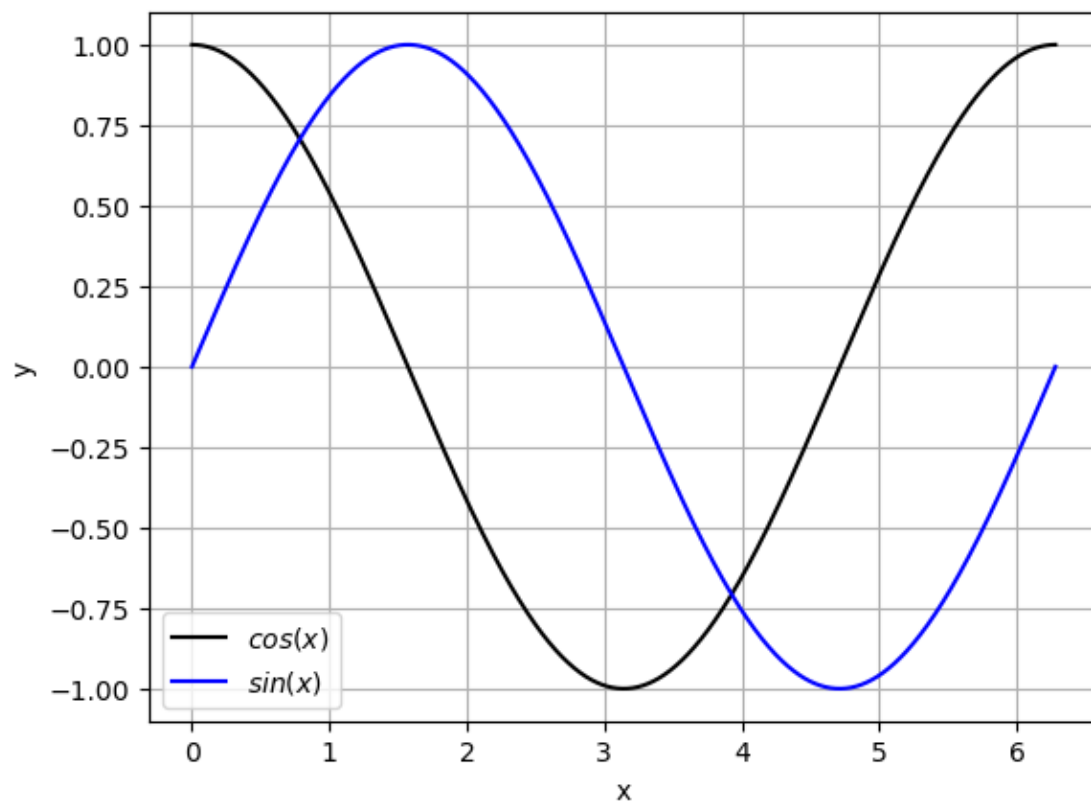
```
[1, 2, 3, 4]
```

```
[1, 2, 4, 3]
```

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[8]: # Graph with matplotlib
```

```
import matplotlib.pyplot as plt  
import numpy as np  
  
x = np.linspace(0, 2*np.pi, 1000)  
y1 = np.cos(x)  
y2 = np.sin(x)  
  
plt.plot(x, y1, '-k', label="$cos(x)$")  
plt.plot(x, y2, '-b', label="$sin(x)$")  
plt.legend()  
plt.xlabel('x')  
plt.ylabel('y')  
plt.grid()
```



[]: